

Week 4 — Gravity: Start-from-Solution Formula Sheet

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Key Results (use as given; no derivations today)

- **Infinite slab (Bouguer slab):** $\Delta g = 2\pi G \Delta \rho t$ (valid when lateral extent $\gg t$ and observation height). Units: SI. Note sign follows $\Delta \rho$.
- **2D cylinder (dike analogue):** use provided response curves and the qualitative rule: *half-width* $\uparrow$ *with depth*; amplitude scales with $\Delta \rho$ and radius; width is insensitive to $\Delta \rho$.
- **Polygonal 2D body (cross-section):** g_z varies piecewise with logarithmic/atan terms; *edge locations* control slope breaks and asymmetry.

Practical cues

- **Width–depth:** half-width/FWHM provides a depth cue (model-dependent). Combine with amplitude and geology for plausibility checks.
- **Superposition:** total field is the sum of fields from all bodies; use normalized kernels, then scale by $\Delta \rho$ and size.
- **Free-air gradient (for practical prep):** target magnitude $\sim 0.3086 \text{ mGal m}^{-1}$ near Earth's surface; compare your measured slope against this value (sign per course convention).

Assumptions & Caveats

- 2D models imply infinite strike; density contrasts are uniform; flat measurement line; terrain effects neglected unless noted.
- Non-uniqueness: different ($\Delta \rho$, depth, size) combinations can fit the same profile; use geological constraints to discriminate.